

Linux Boot Process

Multiple Choice Questions (MCQs)

Instructions: Answer all questions. Each question carries **1 mark**.

Form Link: <https://forms.gle/1WnDSWhda5yW3mZt6>

1. What is the Linux boot process?

- A. Process of installing Linux
- B. Sequence of steps from powering on the system until Linux is ready for users
- C. Process of shutting down Linux
- D. Process of creating users

Answer: B. Sequence of steps from powering on the system until Linux is ready for users

2. Which component runs first when a computer is powered on?

- A. GRUB
- B. Kernel
- C. BIOS / UEFI
- D. systemd

Answer: C. BIOS / UEFI

3. What does BIOS stand for?

- A. Basic Input Output System
- B. Binary Input Output System
- C. Boot Initialization Operating System
- D. Boot Integrated Output System

Answer: A. Basic Input Output System

4. What is the main function of POST?

- A. Start networking
- B. Check hardware components
- C. Start kernel
- D. Start login service

Answer: B. Check hardware components

5. Which hardware component is checked during POST?

- A. RAM
- B. CPU
- C. Keyboard
- D. All of the above

Answer: D. All of the above

6. Which is the modern replacement for BIOS?

- A. GRUB
- B. UEFI
- C. Kernel
- D. systemd

Answer: B. UEFI

7. Which bootloader is most commonly used in Linux?

- A. LILO
- B. GRUB
- C. NTLDR
- D. BOOTMGR

Answer: B. GRUB

8. What does GRUB stand for?

- A. Grand Unified Bootloader
- B. General Runtime Boot Utility
- C. Global Runtime Bootloader
- D. GNU Runtime Boot

Answer: A. Grand Unified Bootloader

9. Where is GRUB stored in legacy BIOS systems?

- A. /boot
- B. MBR
- C. /etc
- D. /home

Answer: B. MBR

10. Which file contains the GRUB configuration?

- A. /etc/grub.conf
- B. /boot/grub/grub.cfg
- C. /boot/kernel.cfg
- D. /etc/default/systemd

Answer: B. /boot/grub/grub.cfg

11. What does the bootloader load into memory?

- A. Login prompt
- B. Linux kernel
- C. Firewall
- D. SSH service

Answer: B. Linux kernel

12. What is the core component of the Linux operating system?

- A. BIOS
- B. Kernel

- C. GRUB
- D. Bash

Answer: B. Kernel

13. Where is the Linux kernel usually stored?

- A. /etc/kernel
- B. /boot/vmlinuz
- C. /usr/kernel
- D. /var/kernel

Answer: B. /boot/vmlinuz

14. What is initramfs?

- A. Kernel module
- B. Temporary root filesystem used during boot
- C. Bootloader configuration
- D. Login service

Answer: B. Temporary root filesystem used during boot

15. What is the process ID of the first user-space process?

- A. 0
- B. 1
- C. 100
- D. 500

Answer: B. 1

16. Which process usually runs as PID 1 in modern Linux systems?

- A. init
- B. bash
- C. systemd
- D. cron

Answer: C. systemd

17. Which command can be used to check the PID 1 process?

- A. ps -p 1
- B. ps -a
- C. top
- D. pid 1

Answer: A. ps -p 1

18. Which systemd target represents command-line mode?

- A. graphical.target
- B. multi-user.target
- C. poweroff.target
- D. rescue.target

Answer: B. multi-user.target

19. Which service is responsible for scheduled jobs?

- A. sshd
- B. crond
- C. httpd
- D. firewalld

Answer: B. crond

20. What is the final stage of the Linux boot process?

- A. Kernel loading
- B. System services start
- C. Login prompt
- D. BIOS execution

Answer: C. Login prompt

